Technical Requirements for a Single-Channel ECG Device

English translation of the supplied Russian task brief

1. Intended use

- Record ECG in one lead.
- Screen heart rhythm.
- Transmit the recording to a physician or telemedicine system.

2. Electrical characteristics

- One channel / one lead.
- Input signal range.
- Sampling rate, for example at least 500 Hz.
- ADC resolution of at least 16 bits.
- Input impedance.
- Common-mode rejection ratio (CMRR).
- Noise level.
- Defibrillator protection.
- Electrode lead-off detection.

3. Filtering

- Baseline wander.
- 50/60 Hz mains interference.
- Muscle interference.
- Ability to disable or adjust filters.

4. Recording

- Recording duration, for example 10-30 seconds.
- Automatic signal quality control.
- Electrode contact indication.
- Heart rate.

5. Software

- Real-time display of the ECG waveform.
- Save each examination.
- PDF / CSV / XML export.
- Patient identification.
- Transfer via USB, Bluetooth, or Wi-Fi.

6. Enclosure and power

- Portable form factor.
- Rechargeable battery.
- USB-C.
- Battery life.
- Weight and dimensions.

7. Telemedicine

- Transmit ECG recordings to a server.
- Associate each recording with a patient ID.
- SCP-ECG / XML / DICOM.
- API for a medical information system or electronic health record.
- Support remote interpretation by a physician or AI.

8. Safety and regulatory requirements

- IEC 60601-1.
- IEC 60601-1-2.
- IEC 60601-2-47 if the device is intended for ambulatory monitoring.
- Requirements for medical-device software.
- Biocompatibility of electrodes and other patient-contacting parts.
- EAEU / Kazakhstan requirements for medical-device registration.

9. Acceptance tests

- Amplitude accuracy.
- Heart-rate accuracy.
- Noise level.
- Interference immunity.
- Filter verification.
- Data-transfer verification.
- Operation when an electrode becomes disconnected.